

TUTORIAL SET

STAT 339

January 2023

1. Find **(i)**. $\langle u, v \rangle$, **(ii)**. $\|\mathbf{u}\|$, **(iii)**. $\|\mathbf{v}\|$, and **(iv)**. $d(u, v)$ for the following given inner product defined on $\mathbb{R}^n$:

(a)

$$\langle u, v \rangle = u \cdot v$$

$$u = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \quad v = \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}$$

(b)

$$\langle u, v \rangle = u_1 v_1 + 2u_2 v_2 + u_3 v_3$$

$$u = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad v = \begin{pmatrix} 2 \\ 5 \\ 2 \end{pmatrix}$$

(c)

$$\langle u, v \rangle = 2u_1 v_1 + 3u_2 v_2 + u_3 v_3$$

$$u = \begin{pmatrix} 8 \\ 0 \\ 8 \end{pmatrix}, \quad v = \begin{pmatrix} 8 \\ 3 \\ 16 \end{pmatrix}$$

(d)

$$u = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, \quad v = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \quad \langle u, v \rangle = u \cdot v$$

(e)

$$u = \begin{pmatrix} 0 \\ 9 \\ 4 \end{pmatrix}, \quad v = \begin{pmatrix} 9 \\ 2 \\ 4 \end{pmatrix}, \quad \langle u, v \rangle = u \cdot v$$

(f)

$$\langle u, v \rangle = u_1 v_1 + 2u_2 v_2, \quad u = \begin{pmatrix} 0 \\ 6 \end{pmatrix}, \quad v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

(g)

$$\langle u, v \rangle = 3u_1v_1 + u_2v_2, \quad u = \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \quad v = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$$

(h)

$$u = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad v = \begin{pmatrix} 7 \\ 9 \end{pmatrix}, \quad \langle u, v \rangle = u \cdot v$$

(i)

$$u = \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \quad v = \begin{pmatrix} 5 \\ 12 \end{pmatrix}, \quad \langle u, v \rangle = u \cdot v$$

(j)

$$\langle u, v \rangle = u \cdot v$$
$$u = \begin{pmatrix} 1 \\ 1 \\ 2 \\ 0 \end{pmatrix}, \quad v = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

2. Use the inner product $\langle A, B \rangle = 2a_{11}b_{11} + a_{12}b_{21} + 2a_{21}b_{12} + a_{22}b_{22}$ to find
(i) $\langle A, B \rangle$, (ii) $\|A\|$, (iii) $\|B\|$, and (iv) θ for the matrices in $M_{2 \times 2}$.

(a)

$$A = \begin{pmatrix} -1 & 3 \\ 4 & -2 \end{pmatrix}, B = \begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix}$$

(b)

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

(c)

$$A = \begin{pmatrix} 1 & -1 \\ 2 & 4 \end{pmatrix}, B = \begin{pmatrix} 0 & 1 \\ -2 & 0 \end{pmatrix}$$

(d)

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, B = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$$

3. Decide whether or not that the set of $n \times n$ real symmetric matrices, denoted by $S_n(\mathbb{R})$, is an inner product space with the inner product defined as $\langle A, B \rangle = \text{tr}(A^T B)$, where $\text{tr}(A)$ denotes the trace of matrix A .
4. Determine if the set of $n \times n$ real skew-symmetric matrices, denoted by $A_n(\mathbb{R})$, is an inner product space with the inner product defined as $\langle A, B \rangle = \text{tr}(A^T B)$, where $\text{tr}(A)$ denotes the trace of matrix A .
5. Let $S = \{u_1, u_2, u_3\}$ be a set of functions in $C[-1, 1]$ such that $u_1(x) = 1$, $u_2(x) = x$, and $u_3(x) = x^2$. Prove or disprove that S is an orthogonal set with respect to the inner product $\langle f, g \rangle = \int_{-1}^1 f(x)g(x)dx$.

6. Let $S = \{u_1, u_2, u_3\}$ be a set of functions in $C[0, \pi]$ such that $u_1(x) = \sin(x)$, $u_2(x) = \cos(x)$, and $u_3(x) = \sin(2x)$. Prove or disprove that S is an orthogonal set with respect to the inner product $\langle f, g \rangle = \int_0^\pi f(x)g(x)dx$.
7. Let $S = \{u_1, u_2, u_3\}$ be a set of functions in $C[-1, 1]$ such that $u_1(x) = \frac{1}{\sqrt{2}}$, $u_2(x) = \sqrt{\frac{3}{2}}x$, and $u_3(x) = \sqrt{\frac{5}{8}}(3x^2 - 1)$. Prove that S is an orthonormal set with respect to the inner product $\langle f, g \rangle = \int_{-1}^1 f(x)g(x)dx$.
8. Let $S = \{u_1, u_2, u_3\}$ be a set of functions in $C[0, \pi]$ such that $u_1(x) = \frac{1}{\sqrt{\pi}}$, $u_2(x) = \sqrt{\frac{2}{\pi}}\sin(x)$, and $u_3(x) = \sqrt{\frac{2}{\pi}}\cos(x)$. Prove that S is an orthonormal set with respect to the inner product $\langle f, g \rangle = \int_0^\pi f(x)g(x)dx$.
9. Determine whether the set $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ is orthogonal, where $\mathbf{w}_1 = (4, -1, 1)$, $\mathbf{w}_2 = (-1, 0, 4)$, and $\mathbf{w}_3 = (-4, -17, -1)$.
10. Find an orthonormal basis for the subspace spanned by the vectors $\mathbf{v}_1 = (1, 1, 1)$ and $\mathbf{v}_2 = (1, 0, -1)$ using the Gram-Schmidt process.
11. Given the vectors $\mathbf{v}_1 = (3, 4, 0, 0)$, $\mathbf{v}_2 = (-1, 1, 0, 0)$, and $\mathbf{v}_3 = (2, 1, 0, -1)$ and $\mathbf{v}_4 = (0, 1, 1, 0)$, apply the Gram-Schmidt process to obtain an orthonormal basis.
12. Find an orthonormal basis for the subspace spanned by the vectors $\mathbf{v}_1 = (1, 0, 1, 0)$, $\mathbf{v}_2 = (0, 1, 0, 1)$, and $\mathbf{v}_3 = (1, 1, 1, 1)$ using the Gram-Schmidt process.
13. Find an orthonormal basis for the subspace spanned by the vectors $\mathbf{v}_1 = (1, 1, 1, 1)$ and $\mathbf{v}_2 = (1, -1, 1, -1)$ using the Gram-Schmidt process.
14. Find an orthonormal basis for the subspace spanned by the vectors $\mathbf{v}_1 = (1, 0, 1, 0)$, $\mathbf{v}_2 = (0, 1, 0, 1)$, $\mathbf{v}_3 = (1, 1, 1, 1)$, and $\mathbf{v}_4 = (1, 0, 0, 0)$ using the Gram-Schmidt process.
15. Find the orthonormal basis for the solution space of the homogeneous system of equations.

(a)

$$2x_1 + x_2 - 6x_3 + 2x_4 = 0$$

$$x_1 + 2x_2 - 3x_3 + 4x_4 = 0$$

$$x_1 + x_2 - 3x_3 + 2x_4 = 0$$

(b)

$$x_1 + x_2 + 7x_4 = 0$$

$$2x_1 + x_2 + 2x_3 + 6x_4 = 0$$

(c)

$$x_1 + x_2 - 3x_3 - 2x_4 = 0$$

$$2x_1 - x_2 - x_4 = 0$$

$$3x_1 + x_2 - 5x_3 - 4x_4 = 0$$

(d)

$$x_1 + 3x_2 - 3x_3 = 0$$

(e)

$$x_1 - 2x_2 + x_3 = 0$$